

Comprehensive Analysis of WIS2 Monitoring Alerts: Infrastructure Diagnostics, Schema Validation, and Systemic Telemetry

1. Introduction to the WIS2 Observability Framework

The transition of the World Meteorological Organization (WMO) to the Information System 2.0 (WIS2) represents a fundamental architectural evolution. Moving away from the legacy, point-to-point file exchange mechanisms characteristic of the Global Telecommunication System (GTS), WIS2 embraces a highly available, globally distributed Publish/Subscribe (PubSub) architecture.¹ This modern paradigm relies upon a synergistic network of core infrastructure components known as Global Services. These encompass Global Brokers for message routing, Global Caches for data payload hosting, and Global Discovery Catalogues (GDCs) for centralized metadata querying and discoverability.² Operating effectively within this ecosystem requires stringent adherence to performance metrics, including a mandate for Global Brokers to achieve at least 99.5% service availability, process a minimum of 10,000 messages per second, and seamlessly support hundreds of concurrent WIS2 Nodes.² To maintain this unprecedented scale and reliability, comprehensive system observability is paramount. This observability is formalized through the WIS2 Monitoring Events (WME) specification.¹ The WME specification mandates the use of the CloudEvents standard, encoded in JavaScript Object Notation (JSON), to construct standardized telemetry messages.¹ These messages are broadcast over a dedicated Message Queuing Telemetry Transport (MQTT) topic hierarchy—specifically `monitor/a/wis2/#`—which allows Global Services and specialized monitoring agents to continuously ingest state changes, error reports, and system health metrics.¹ Through this mechanism, the WMO has established a real-time diagnostic nervous system capable of identifying everything from localized network timeouts to global schema validation failures.¹

This report provides an exhaustive, forensic analysis of the WIS2 monitoring alerting ecosystem. It extracts, categorizes, and deeply analyzes currently captured telemetry from active monitoring environments, cross-referencing these empirical findings with normative WMO architectural schemas to identify theoretical and uncaptured alerts.¹ Furthermore, the analysis details the specific trigger conditions, severity categorizations, underlying technical root causes, downstream impacts on meteorological data flow, and actionable remediation strategies for every identified alert class. By weaving together the technical mechanisms of MQTT, HTTP protocol dynamics, and JSON schema validation, this report serves as the definitive reference for monitoring agents and infrastructure operators responsible for the resilience of the WIS2 global network.

2. The Architectural Anatomy of WIS2 Monitoring Events

Before delving into the specific alert classifications, it is essential to understand the structural composition and dissemination mechanics of the monitoring events themselves. WIS2 Monitoring Event Messages (WMEM) are strictly bounded by the <http://wis.wmo.int/spec/wme/1/req/monitoring-event-message-core> requirement class, ensuring that all emitted telemetry adheres to a predictable and machine-actionable format.¹ The metadata payloads within these JSON structures are designed to be compact, with a strict requirement not to exceed 8192 bytes, ensuring minimal overhead on the message brokers.¹ Every event is uniquely tracked via an id property utilizing a Universally Unique Identifier (UUIDv4), preventing message collision and enabling precise chronological tracing across distributed log aggregators.¹ The type property is arguably the most critical classification mechanism, utilizing a reverse DNS string format (such as `int.wmo.wis.wme.event.wcmp2.ets` or `int.wmo.wis.wme.wnm.validation.schema`) to explicitly define the nature of the anomaly.¹ Furthermore, event routing is dependent on the source and subject properties, which utilize center identifiers defined by the WIS2 Topic Hierarchy.¹ The source denotes the specific agent or monitoring service that detected the anomaly, while the subject identifies the failing or notifying entity.¹ To facilitate automated triage and response escalation, every event includes a `data.severity` property. This property is strictly enumerated into `DEBUG`, `INFO`, `WARNING`, `ERROR`, and `CRITICAL` levels.¹ Finally, `data.content` and `data.links` provide human-readable diagnostic summaries and machine-actionable endpoints, respectively, allowing operators to rapidly query related metadata records or external documentation associated with the failure.¹ The ingestion of this telemetry is achieved through MQTT wildcard subscriptions. Global Monitors and regional observability platforms subscribe to channels such as `monitor/a/wis2/+` or `monitor/a/wis2/ca-eccc-msc/#`, allowing them to construct a continuous, real-time topology of network health without generating polling overhead on the individual WIS2 Nodes.¹

3. Exhaustive Analysis: Node Connectivity and Broker Disconnections

A foundational requirement of the WIS2 architecture is that every data-producing Node must maintain continuous, concurrent MQTT connections to multiple Global Brokers.² This redundant mesh ensures that if one broker experiences a localized outage, the WIS2 Notification Messages (WNM) are still successfully routed through alternative paths. Telemetry data from active WIS2 environments captures distinct degradation states within this connectivity mesh, each carrying different systemic implications.⁴

Event Title	Severity	Event Type Code	Core System Impact

Disconnected WIS2 Node from one Global Broker	WARNING	int.wmo.wis.wme.event	Redundancy degradation; no data loss.
Disconnected WIS2 Node from multiple Global Brokers	ERROR	int.wmo.wis.wme.event	Severe redundancy loss; elevated latency risk.
Disconnected WIS2 Node from all Global Brokers	CRITICAL	int.wmo.wis.wme.event	Complete notification blackout; global data isolation.
Target is down (e.g., Target wis2.ncm.gov.sa:443 is down)	CRITICAL	int.wmo.wis.wme.event	Node endpoint entirely unreachable for monitoring.

3.1 Degradation of the Broker Redundancy Mesh

The Disconnected WIS2 Node from one Global Broker alert operates at a WARNING severity level and serves as the earliest indicator of network instability.⁴ The trigger mechanism for this alert is initiated when a Global Monitor detects that the continuous heartbeat, or the underlying TCP socket state, of a specific WIS2 Node has dropped from one of its configured target Global Brokers while remaining active on the others. The underlying technical causes are typically localized. These include transient network routing anomalies across specific international submarine cables, firewall misconfigurations where a network administrator inadvertently blocks outbound traffic on port 8883 (MQTTS) or port 443 (WSS) to a specific broker's IP address block, or localized Transport Layer Security (TLS) handshake failures due to geographic load-balancing discrepancies.² The immediate impact on meteorological data flow is negligible, as the inherent redundancy of WIS2 ensures that WNMs are successfully published via the remaining connected brokers.² However, actionable responses require network engineering teams to review outbound firewall rules and perform diagnostic packet tracing to the specific disconnected broker to prevent further degradation.

When the alert escalates to Disconnected WIS2 Node from multiple Global Brokers, categorized as an ERROR, the threat profile changes significantly.⁴ This is triggered by simultaneous connection drops across two or more, but crucially not all, Global Brokers. The technical causes for this phenomenon are usually indicative of broader infrastructural issues, such as regional internet backhaul outages, Border Gateway Protocol (BGP) route flapping that affects multiple international transit paths, or resource starvation within the Node's message

queue client, causing it to aggressively terminate parallel connection threads. While some redundancy remains intact, the data flow impact is characterized by increased notification latency, and the Node operates precariously close to a total blackout. Monitoring agents must immediately escalate this event to the Node operator's local network engineering team to inspect edge router stability, outbound connection limits, and the CPU utilization of the publishing daemon.

3.2 Total Isolation and Endpoint Failure

The most severe connectivity anomaly is the Disconnected WIS2 Node from all Global Brokers event, classified as CRITICAL.⁴ The trigger mechanism occurs when the Global Monitor registers absolute silence from the Node; zero active MQTT connections exist across the entire Global Broker mesh.⁴ The technical causes represent catastrophic failures at the Node level. These may include total facility power loss, complete physical isolation from Internet Service Providers (ISPs), the expiration of top-level TLS certificates preventing any encrypted handshake, or fatal runtime crashes within the underlying messaging software, such as wis2box or a proprietary equivalent. The impact on data flow is absolute and devastating for the subject Node. Because no WNM's can be published, the global community remains entirely unaware of any new meteorological data generated by the Node, effectively erasing its real-time scientific contribution. Actionable responses dictate immediate, direct out-of-band communication with the Node administrator, necessitating emergency restarts of the publishing infrastructure and physical verification of core internet connectivity.

Closely related to total broker disconnection is the Target is down alert, also classified as CRITICAL.⁴ For example, the telemetry highlights events where monitoring probes failed to reach www.wis-jma.go.jp:443 or wis2.ncm.gov.sa:443.⁴ This alert is triggered when active health checks—typically continuous HTTP GET requests or ICMP pings originating from the Global Monitor—fail to receive any response. The technical causes generally involve edge network failures, such as reverse proxy crashes (e.g., Nginx or HAProxy dropping offline), distributed denial of service (DDoS) mitigation hardware aggressively blackholing traffic, or total server hardware failure. The impact extends beyond notification failure; it indicates that even if notifications were somehow queued, the endpoint hosting the actual data payload or metadata services is completely inaccessible. Actionable responses require local system administrators to verify the status of edge load balancers, web server daemons, and physical server availability.

4. Exhaustive Analysis: Global Cache and Data Flow Disruptions

While Global Brokers manage the signaling and notification pathways, Global Caches are responsible for the heavy lifting of payload distribution. When a WNM announces new data, Global Caches download the data payload from the origin WIS2 Node's HTTP server and cache it locally.² End-users then download the data from the Cache, thereby shielding the origin Node from overwhelming concurrent bandwidth requests.² Anomalies in this domain signify a critical breakdown between the notification phase and the actual data retrieval phase, leading to

phenomena where data is announced but physically unobtainable.

Event Title	Severity	Event Type Code	Core System Impact
No cache is receiving data from that center over the last 6 hours	ERROR	int.wmo.wis.wme.ev ent.item.download	Global "ghosting" effect; data is announced but inaccessible.
No data is received from that center over the last 6 hours	WARNING	int.wmo.wis.wme.ev ent.item.download	Potential localized cache failure or slow data production.
The Global cache is not receiving any data since 1 hour	CRITICAL	int.wmo.wis.wme.ev ent.item.download	Cache-wide ingestion failure; broad data staleness for subscribers.
Global cache has > 10 download errors from center in 1h over 6h	WARNING/ERROR	int.wmo.wis.wme.ev ent.item.download	High friction data fetching; partial payload loss and latency.
Global Cache cannot connect to > 10 (or 20) dataservers	ERROR/CRITICAL	int.wmo.wis.wme.ev ent.item.download	Cache egress failure; inability to populate cache inventory.

4.1 Ingestion Failures and the "Ghosting" Phenomenon

The No cache is receiving data from that center over the last 6 hours alert, categorized as an ERROR, highlights a profound systemic dissonance.⁴ The trigger mechanism relies on a time-series aggregation confirming that for a continuous six-hour window, WNM's are being successfully published by the origin Node, but zero bytes of the associated payload data have been successfully ingested by any Global Cache.⁴ The underlying technical causes usually point to the origin Node publishing valid MQTT notifications, but the canonical HTTP links embedded within those WNM's point to an inaccessible data server. This can occur if the server is hidden behind a private Network Address Translation (NAT) firewall, if the DNS records for the data

server have been misconfigured or have not propagated globally, or if the HTTP daemon serving the physical files has crashed while the MQTT publishing script continues to run blindly. The impact on data flow results in a severe "ghosting" effect. Automated subscriber pipelines receive notifications of new weather data, but their subsequent programmatic attempts to download the payloads encounter dead links, rendering the data entirely inaccessible and causing widespread script timeouts. Actionable responses require the Node operator to verify the HTTP/HTTPS reachability of the canonical URL provided in their WNM from external, global networks, ensuring the data storage endpoint is genuinely public-facing.

A localized variant of this issue is the The Global cache is not receiving any data since 1 hour alert, which carries a CRITICAL severity.⁴ Unlike the previous alert which tracks a specific origin center, this alert is triggered when a specific Global Cache reports a sudden, absolute drop in its overall incoming data ingestion rate across all global centers for sixty consecutive minutes.⁴ The technical causes invariably point to an internal, catastrophic failure within the Global Cache's own infrastructure. This could stem from exhausted disk space on the Cache's storage arrays, out-of-memory (OOM) errors that have aggressively killed the worker processes responsible for executing the HTTP downloads, or a scenario where the Cache's outbound IP address block has been inadvertently blacklisted by origin centers due to perceived aggressive polling behavior. The impact on data flow is massive for any end-user or institution relying on this specific Cache as their primary data source; they will be served increasingly stale data or begin encountering HTTP 500 and 503 errors. Actionable responses mandate that Cache operators immediately interrogate their infrastructure dashboards to check storage volume metrics, restart containerized worker nodes, and inspect local networking interfaces for packet loss.

4.2 Friction and Rate Limiting Anomalies

Not all caching failures are absolute; many represent high-friction states that cause partial data loss. This is captured by alerts such as The Global cache has more than 10 download errors from center in 1 hour over the last 6 hours, which fluctuate between WARNING and ERROR severities depending on the breadth of the Caches affected.⁴ The trigger mechanism utilizes a rolling window function that detects persistent, repeating HTTP 4xx or 5xx error codes when Caches attempt to fetch data payloads from a specific Node's data server.⁴

The underlying technical causes for this friction are multifaceted. The Node's data server may be applying overly aggressive rate-limiting configurations, resulting in a barrage of HTTP 429 Too Many Requests responses. Alternatively, the Node's server might be under-resourced, experiencing intermittent gateway timeouts (HTTP 504) as it struggles to read large datasets from its disk into memory. Another common cause is rapid data deprecation, where the Cache receives an HTTP 404 Not Found error because the Node's physical data files are being deleted by local cleanup scripts faster than the distributed Caches can retrieve them. The impact on data flow involves partial data loss and high latency. While some data is successfully cached, a significant percentage is permanently lost during the error windows, leading to fragmented and incomplete meteorological datasets downstream. Actionable responses require the origin Node operator to meticulously tune their web server's rate-limiting logic to accommodate Cache harvesting, scale their HTTP server resources to handle burst requests, or extend the

persistence window of their physical data files before executing deletion routines. Similarly, alerts tracking Cache egress limitations—such as The Global Cache can not connect to more than 10 dataservers in the last hour (ERROR) or the escalation to 20 dataservers (CRITICAL)—indicate that the Cache is losing its ability to reach the outside world.⁴ Triggered by aggregated HTTP connection failures (timeouts, connection refused states), the technical causes suggest compromised network egress at the Cache facility. This could be due to upstream ISP throttling, internal DNS resolution failures preventing the Cache from finding the IP addresses of the origin servers, or an overloaded outbound NAT gateway exhausting its available ephemeral ports. Actionable responses necessitate immediate investigation of the Cache's internal DNS resolvers and outbound routing appliances.

5. Exhaustive Analysis: Protocol, Endpoint, and Network Layer Terminations

Beneath the architectural logic of Caches and Brokers lies the fundamental network layer. Monitoring agents in the WIS2 ecosystem continually scrape Global Services for performance telemetry using tools like Prometheus. When the foundational protocols fail, the observability plane goes blind. These alerts capture deep protocol and endpoint anomalies.⁴

5.1 Connection Timeouts and Deadlines

The Timeout: context deadline exceeded alert, categorized as an ERROR, is one of the most pervasive indicators of system strain. Target examples from empirical data include core components like `globalbroker.inmet.gov.br`, `wis2.ncm.gov.sa`, and `wis2-gdc.weather.gc.ca`.⁴ The trigger mechanism occurs when a Global Monitor initiates an HTTP GET request (often targeting a `/metrics` or `/prometheus/federate` endpoint), but the TCP connection or the subsequent HTTP response is not completed within the strict timeout window defined by the application's programming logic (specifically, the Golang `context.Context` deadline utilized by many modern monitoring tools).⁴

The underlying technical causes for this timeout are highly indicative of resource exhaustion. The target server may be experiencing extreme CPU load, causing the application threads to stall and delay the generation of the requested metrics. In scenarios involving Global Discovery Catalogues, backend database locking (e.g., within PostgreSQL or Elasticsearch clusters) can severely delay the assembly of the metrics payload. Additionally, a stateful firewall located in front of the target may be silently dropping packets without returning a TCP RST (Reset) flag, forcing the monitor's connection to hang until the deadline expires.

The systemic impact on data flow is insidious. While the actual meteorological data flow and MQTT routing *might* still be functioning normally, the monitoring plane is effectively blinded. This lack of visibility raises the critical risk of silent, cascading failures occurring without operator awareness. Actionable responses require monitoring engineers to temporarily increase timeout thresholds in scraper configurations to allow slow targets to respond, while simultaneously requiring the target operators to optimize the computational performance of their `/metrics` endpoints and investigate CPU load averages on the affected host machines.

5.2 Abrupt Terminations and Access Denials

The Network Termination: unexpected EOF alert, classified as an ERROR, represents a violent interruption of the communication channel.⁴ Observed on targets like gc.wis.cma.cn, the trigger mechanism occurs when the underlying TCP connection is abruptly severed by the remote server before the complete HTTP payload is transmitted to the monitoring agent.⁴ The technical causes are typically isolated to the edge network infrastructure. The reverse proxy acting as the gateway for the target node may have crashed mid-stream due to a segmentation fault. Alternatively, an upstream load balancer may have prematurely killed the connection due to aggressive, misconfigured keep-alive timeouts, or the TLS session may have attempted an aggressive renegotiation that failed, resulting in a dropped socket. The impact on data flow involves metric loss and potential instability in actual data fetching, particularly if the same reverse proxy is responsible for handling both telemetry requests and WNM payload downloads. Actionable responses require target administrators to inspect their web server error logs for application panics and review proxy timeout configurations. Conversely, the HTTP Error: 403 Forbidden alert (e.g., observed on www.wis-jma.go.jp) indicates a successful network connection but a deliberate rejection by the application layer.⁴ This ERROR is triggered when the target server explicitly returns an HTTP 403 status code. The technical causes almost exclusively point to security infrastructure interference. A Web Application Firewall (WAF) may possess rules that falsely identify the Global Monitor's User-Agent string or polling frequency as malicious, effectively blacklisting the monitor's IP address. It can also stem from incorrectly applied cross-origin resource sharing (CORS) policies. The impact is a complete inability for the global network to verify the node's health. Actionable responses dictate that the target operator must update their WAF Access Control Lists (ACLs) to explicitly whitelist the known IP ranges of the WIS2 Global Monitors.

6. Exhaustive Analysis: Schema and Metadata Validation Integrity

The operational success of WIS2 relies not just on moving bytes from point A to point B, but on ensuring that those bytes are universally comprehensible and discoverable. Data discoverability is strictly governed by compliance with the WMO Core Metadata Profile (WCMP2) and the structural integrity of the WIS2 Notification Message (WNM) schema.¹ Alerts in this domain are driven by highly automated JSON schema validators and Executable Test Suites (ETS) managed by Global Discovery Catalogues.¹

Event Title	Severity	Event Type Code	Core System Impact
WIS2 Notification Message not compliant with	ERROR	int.wmo.wis.wme.wnm.validation.schema	Message quarantine; downstream

schema			parsing panics.
Missing Metadata record for data announced in WNM	ERROR	int.wmo.wis.wme.wnm.validation.meta data	Data is "dark" and completely undiscoverable via GDC.
WCMP2 Access Failure	ERROR	int.wmo.wis.wme.event	GDC cannot retrieve mandatory discovery records.
WCMP2 ETS Report (Schema constraint failures)	INFO / ERROR	int.wmo.wis.wme.event.wcmp2.ets	Rejection of non-compliant metadata by catalogues.
WCMP2 KPI Report Degradation	WARNING	int.wmo.wis.wme.event.wcmp2.kpi	Severe penalization in global search rankings.

6.1 Payload and Structural Validation

The WIS2 Notification Message not compliant with the defined schema alert represents a fundamental breakdown in message construction.⁴ Classified as an ERROR, the trigger mechanism occurs the moment a Global Broker or monitoring tool evaluates the JSON payload of an incoming WNM against the normative `wis2-notification-message-bundled.json` schema and detects a structural violation.² Empirical data provides specific examples of this, such as when the data field `properties.content.size` exceeds its configured maximum limit (e.g., "should be <= 4096").⁴

The underlying technical causes are typically deeply rooted in the software bugs of the Node's message generation logic. This includes scripts that fail to properly sanitize inputs, manual formatting errors when constructing static messages, or failures to correctly escape complex JSON strings before transmission. The impact on data flow is immediate and systemic. Global Brokers tasked with maintaining network hygiene may quarantine or outright drop the malformed message, preventing it from propagating.² Furthermore, downstream subscribers executing strict parsing routines within their NWP ingestion engines will experience fatal application crashes if they attempt to deserialize the invalid JSON. Actionable responses demand that Node developers leverage available tooling, such as `pywis-pubsub` or the `check-jsonschema` utility, to pre-validate their generated payloads locally before initiating any MQTT publication.²

6.2 The Discoverability and "Dark Data" Threat

A perfectly formed WNM is useless if the global community cannot search for and understand the data it represents. The Missing Metadata record for data announced in WIS2 Notification Message alert captures this critical discrepancy.⁴ Operating as an ERROR, the trigger mechanism is invoked when a Global Broker receives a WNM containing a specific metadata_id (e.g., urn:wmo:md:uk-metoffice-wmc:weather.prediction...), but cross-referencing this ID with the Global Discovery Catalogue reveals that the record simply does not exist.²

The technical causes for this phenomenon involve asynchronous operational workflows. Node operators frequently deploy new data generation pipelines and configure their MQTT publishers to announce the data, but neglect to publish the accompanying WCMP2 discovery record to the GDC. Alternatively, the operator may have attempted to publish the WCMP2 record, but the GDC rejected it during its own ingestion validation phase. The systemic impact is the generation of "dark data." Subscribers who are explicitly listening to the specific topic will receive the data, but no new end-users or researchers can discover the dataset via GDC web interfaces or APIs because the descriptive context is missing. This violates the core discovery principles of WIS2. Actionable responses require the Node operator to immediately generate a compliant WCMP2 record utilizing the exact metadata_id referenced in the WNM and push it to the catalog system.

Furthermore, if the GDC attempts to fetch a referenced metadata record and encounters a WCMP2 access failure (an HTTP 404 Not Found error from the Node's metadata repository), an ERROR event is generated.⁴ This indicates that the metadata endpoint URL was changed without notice, the file was accidentally deleted, or directory permissions were improperly modified. Operators must immediately restore the WCMP2 JSON file to the correct URI path to restore discoverability.

6.3 Executable Test Suites and Quality Scoring

To maintain absolute standardization, GDCs deploy automated validation engines known as Executable Test Suites (ETS).¹ The resulting WCMP2 ETS report events, generally emitted as INFO telemetry (though failing scores escalate to errors), provide a granular, JSON-formatted array of passed and failed constraints.¹ The trigger mechanism is the routine evaluation of published WCMP2 records. The technical logic tests for strict adherence to constraints such as geospatial bounding box formatting, temporal ISO 8601 syntax, and the inclusion of mandatory data policy links.⁴ If the ETS evaluation results in a FAILED code, the GDC outright refuses to ingest the record, rendering the associated data undiscoverable.⁵ Node operators must review the specific failed schema constraints highlighted in the telemetry and republish the corrected record.

Moving beyond simple binary compliance, the WIS2 framework also theorizes WCMP2 KPI Report events, utilizing the int.wmo.wis.wme.event.wcmp2.kpi type code.¹ While not explicitly captured as an error in the sample CSV, it is a critical component of the monitoring architecture. Key Performance Indicators (KPIs) assess the qualitative value of the metadata—for instance, evaluating the depth and descriptiveness of the abstract, or the

precision of the spatial bounding boxes.⁵ If a record passes ETS but scores abysmally on the KPI index, a WARNING monitoring event is generated. This alerts the data provider that while their data is technically valid, its discoverability ranking within the GDC search algorithms will be heavily penalized until the descriptive metadata is enriched.

7. Systemic Telemetry: Lifecycle Notices and Latent Threats

A holistic observability framework does not solely track failures; it must also account for administrative events and theoretical threats outlined within the architecture's foundational documentation.

7.1 Scheduled Maintenance and Administrative Silencing

Not all alerts imply systemic failure; some are vital for incident management and downstream pipeline planning. Telemetry captures numerous Scheduled Maintenance Notices (e.g., "A scheduled maintenance service for CMA Global Monitor", "Maintenance Service for CMA Global Broker").⁴ Though these are technically administrative notices, they are frequently flagged with a CRITICAL severity level because they dictate the total, albeit planned, unavailability of a core system component.⁴

The trigger mechanism for these events is the manual or automated generation of a planned outage template (often classified as `int.wmo.wis.wme.event.notice`), which is broadcasted globally across the MQTT monitoring topics.¹ The underlying technical causes for such maintenance windows include SSL certificate rotations, physical infrastructure migrations, emergency hardware upgrades, and significant software transitions, such as upgrading a Global Broker's engine to support MQTT v5.0.² The impact on data flow is strictly temporary but highly disruptive if unmanaged. Nodes explicitly warn that data feeds or caching capabilities will be suspended during specified UTC windows. In the case of Global Broker maintenance, the architecture's redundancy ensures alternative brokers absorb the traffic.⁴ However, for subscribers relying entirely on a specific Node that is undergoing maintenance, actionable responses dictate that ingestion pipelines must be paused, or failover scripts must be engaged to source data from alternative regional providers until the maintenance window closes.

7.2 Broker Routing Integrity and Anti-Loop Detection

While extracting active errors provides insight into common failures, analyzing the normative WMO specifications reveals latent alerts that are critical to systemic survival but missing from the current sample set. Foremost among these are alerts related to Broker routing loops.² Global Brokers are explicitly mandated to subscribe to each other to ensure redundant and reliable transmission across the globe.² However, this interconnected mesh introduces the risk of infinite message loops. If a poorly configured Node recursively republishes the same data, or if the brokers fail to track message origins, an infinite broadcast storm can occur. To prevent this, brokers implement an Anti-loop mechanism that tracks message UUIDs and suppresses duplicates.²

The theorized alert for this scenario—an Anti-loop threshold exceeded or Duplicate message detection event (WARNING or ERROR)—is triggered when the duplicate detection heuristic flags an exceptionally high rate of identical UUIDs within a sub-second timeframe. The technical cause is a failure in the routing logic or a malfunctioning Node entering an infinite retry loop. The systemic impact of an unchecked loop is catastrophic: exponential network bandwidth consumption, broker CPU exhaustion, and the eventual crash of the entire PubSub mesh. Monitoring agents must be vigilant for these alerts to immediately quarantine the offending Node's topic stream.

7.3 Global Discovery Catalogue Latency

Another critical theorized alert explicitly referenced in the WME core documentation is the detection of a Global Discovery Catalogue metadata archive being older than 24 hours.¹ Operating at a WARNING or ERROR severity, this alert is triggered when a GDC's internal synchronization process (e.g., a scheduled harvester pulling records via OGC API endpoints) fails to update its index for a full day. The technical causes typically involve backend database hanging (e.g., a locked indexing queue) or the harvester scripts crashing silently. The systemic impact is that the GDC becomes entirely "stale." Users querying the catalogue for newly registered data nodes will receive outdated or null results, directly undermining the WIS2 principle of rapid global data discovery. Actionable responses require GDC administrators to manually force index rebuilds and inspect harvester application logs.

7.4 The Sensor Centre Paradigm and Performance Thresholds

The most advanced aspect of the WIS2 observability framework is the conceptualization and deployment of "Sensor Centres".¹ Spearheaded by the WMO's Expert Team on WIS2 Implementation and Transition (ET-W2IT), Sensor Centres act as independent, objective auditing entities distributed across multiple WMO Regions.³ These encompass specific deployments such as the Sensor Centre Global Broker (SCGB), Sensor Centre Global Cache (SCGC), and Sensor Centre Global Discovery Catalogue (SCGDC).³

These Sensor Centres are explicitly designed to generate WME alerts when operational Key Performance Indicators (KPIs) and Service Level Agreements (SLAs) are breached. For example, if a Global Broker fails to meet the stringent architectural requirement of processing a minimum of 10,000 messages per second, or if its subscriber capacity breaches 1,000 active connections without dynamically scaling, the Sensor Centre's localized OpenMetrics and Prometheus AlertManager scripts will independently generate and push a CRITICAL WME alert to the global network.²

The technical cause of such alerts is pure computational or bandwidth exhaustion at the target Global Service facility. The systemic impact of capturing these metrics via independent Sensor Centres allows for the proactive identification of service degradation long before it cascades into an outright failure or a total broker disconnection. It provides the global monitoring community with undeniable, objective telemetry to enforce operational standards.

Furthermore, the Sensor Centre framework is inherently extensible; regional authorities can deploy bespoke Sensor Centres to validate local regulatory compliance (e.g., validating Global Basic Observing Network (GBON) or Regional Basic Observing Network (RBON) data flow),

integrating their findings seamlessly into the overarching WME alerting topic.³

8. Actionable Response Framework for Monitoring Agents

The ultimate value of the WIS2 Monitoring Events specification is not merely in the generation of complex JSON payloads, but in the operationalization of that telemetry into rapid incident response. To standardize recovery, monitoring agents, Global Information System Centres (GISCs), and local Node operators must adhere to a strict response matrix governed by the event's data.severity classification.

Severity Level	Threat Profile & Example Alerts	Immediate Technical Action	Escalation & Remediation Strategy
DEBUG / INFO	Non-disruptive. Routine telemetry (e.g., WCMP2 ETS Passed, standard system heartbeats). ¹	No manual intervention. Ingest to time-series databases for long-term SLA reporting.	Used by Sensor Centres to tune KPI algorithms and adjust baseline anomaly detection thresholds. ³
WARNING	Degradation of redundancy or creeping friction. (e.g., Disconnected from one Global Broker, KPI score dipping). ⁴	Trigger automated localized network diagnostics (traceroute, ping). Verify local TLS certificates. ²	Log a non-urgent ticket for next-business-day review. Investigate underlying ISP jitter or minor JSON metadata omissions.
ERROR	Isolated failure affecting specific data sets. (e.g., WNM schema failure, 403 Forbidden, Missing Metadata). ¹	Quarantine the malformed payload. Suspend downstream fetching from the specific failing center to prevent connection exhaustion.	Alert data provider immediately. Require manual schema correction (check-jsonschema) or WAF ACL whitelisting before unblocking. ⁴

CRITICAL	Systemic or total blackout. (e.g., Disconnected from <i>all</i> brokers, Cache 0 bytes ingested, Node target Down). ¹	Engage emergency fallback. Reroute subscriber scripts to regional backup Caches and alternative brokers.	Page standby operations engineers. Escalate to hardware/ISP level. Extensive postmortem analysis required upon restoration. ⁴
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9. Conclusion

The WIS2 Monitoring Events (WME) framework represents a highly sophisticated, CloudEvents-driven observability layer that is indispensable for the maintenance and scalability of global meteorological data exchange. The architectural transition from static file transfers to a dynamic, asynchronous PubSub mesh introduces immense complexity; while it decouples producers and consumers, it simultaneously introduces the risk of silent, localized failures that can rapidly cascade into global data blackouts.

As evidenced by the exhaustive analysis of empirical telemetry, WIS2 alerts provide unparalleled, granular diagnostics. These range from the detection of critical physical infrastructure failures (Node disconnections, HTTP target timeouts) to nuanced, deep-packet inspections of JSON schema violations and metadata non-compliance. By deeply categorizing these alerts into connectivity matrices, data flow friction points, endpoint protocol anomalies, and strict schema validation verticals, it becomes demonstrably clear that maintaining the WIS2 infrastructure requires a proactive, highly automated approach to system administration. Furthermore, the WMO's forward-looking integration of independent Sensor Centres and automated Executable Test Suites (ETS) ensures a continuous, objective audit of the network's health. This guarantees that not only is the uptime of the physical network relentlessly tracked, but the integrity, compliance, and ultimate scientific discoverability of the meteorological data itself remain entirely uncompromised. For infrastructure operators and monitoring agents across the globe, the continuous ingestion, automated parsing, and rapid operationalization of these WME alerts are paramount to securing the fault-tolerant future of the WMO Information System 2.0.

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